Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Production ML, Offline Evaluation, Data Pipelines | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: Senior ML/AI Engineer – Canadian Tire Corporation

Dear Hiring Manager,

I am applying for the Senior ML/AI Engineer role in Toronto at Canadian Tire Corporation. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I own ML systems end-to-end: data, training, evaluation, release and operational support. Your role centres on making offline evaluation predict online results for personalized loyalty offers, and on a platform where retraining is routine. Careful evaluation and reproducible pipelines are the core of how I work.

Three parts of your posting match my work directly. For production ML and pipelines, I led ML and computer vision deployments at MAS Holdings that raised operational efficiency by up to 300%, built real-time ingestion and ETL pipelines for millions of IoT events a day, and set up code review and CI/CD while mentoring engineers. For evaluation, I design benchmark suites with baselines and ablations; my published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline (F1 0.76 vs 0.65, Journal of the Audio Engineering Society, 2026), and I released four open datasets on Zenodo. For orchestration, I run Airflow (MWAA), Docker and Snowflake pipelines on AWS and use coding agents in my daily workflow.

I should be direct about the gaps. I have not built a production recommender or large-scale ranking system, and I have not used PySpark at hundreds of millions of rows, LightGBM or XGBoost in production, Vertex AI or BigQuery, or Terraform. My cloud ML experience is AWS SageMaker, and my deep learning work is in TensorFlow, Keras and PyTorch. I have built the surrounding pipelines and evaluation harnesses on other stacks, and I expect to learn two-tower retrieval, learning-to-rank and Spark profiling quickly, starting from your loyalty data. I would aim to contribute early to the evaluation framework and the reproducibility of training pipelines.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, available immediately, and comfortable with the hybrid arrangement in Toronto. I would welcome the opportunity to discuss how my experience can support your team at Canadian Tire Corporation. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD